

Version: 1.0

Technical Specification

**MODEL NO: VB3300-KHB
(ES108FC2)**

The content of this information is subject to be changed without notice.
Please contact E Ink or its agent for further information.

☐ Customer's Confirmation

Customer _____

Date _____

By _____

☐ E Ink's Confirmation

Approved By 林建陽
Confirmed By 呂嘉祐
Prepared By 丁冠福

Revision History

Rev.	Issued Date	Revised Contents
V1.0	20200326	Create

TECHNICAL SPECIFICATION

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1. General Description

VB3300-KHB is a reflective electrophoretic E Ink® technology display module based on plastic active matrix TFT substrate. It has 10.8" active area with 1920(H) x 1080(V) pixels, the display is capable to display images at 2-16 gray levels (1-4 bits) depending on the display controller and the associated waveform file it used.

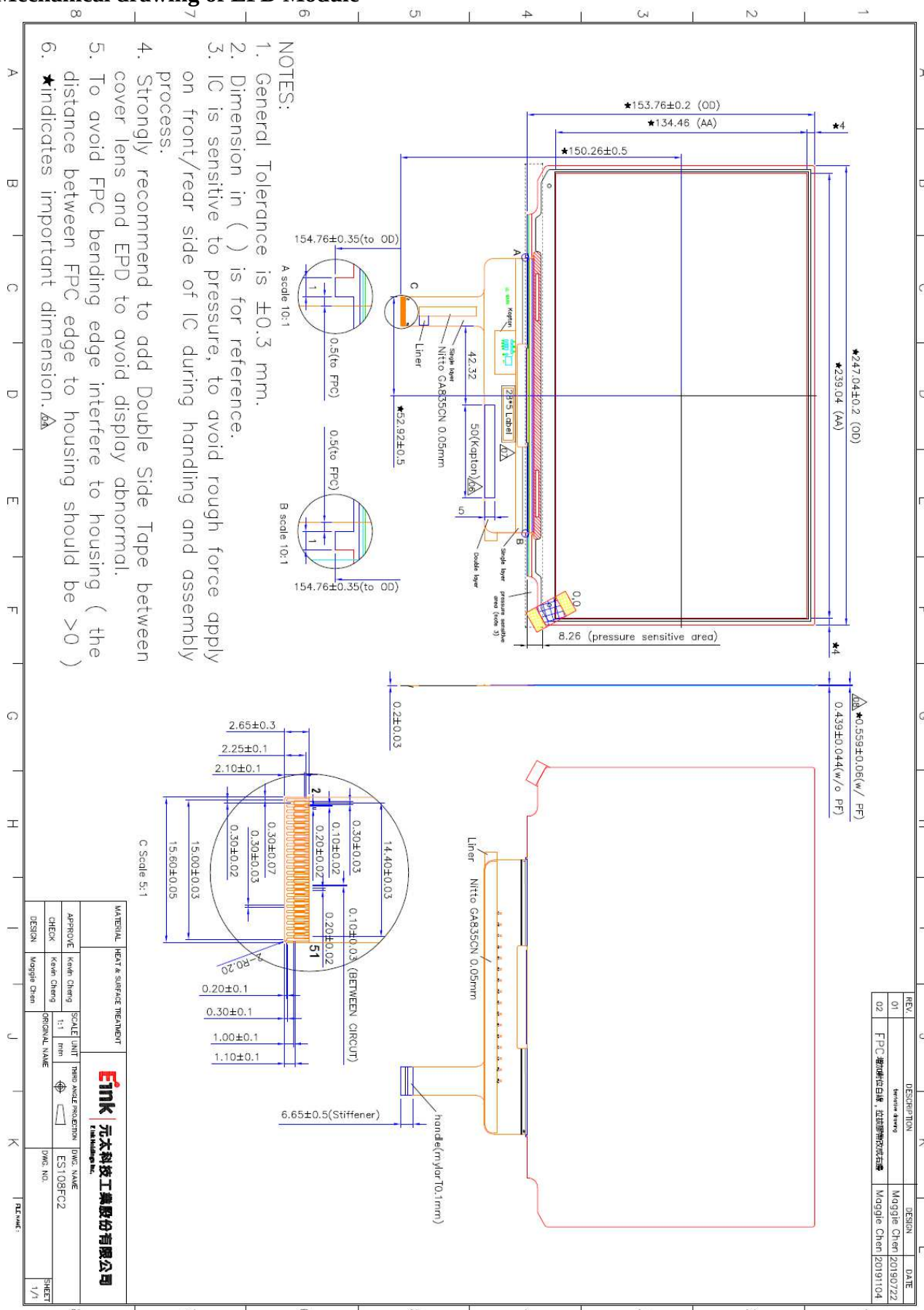
2. Features

- High contrast reflective/electrophoretic technology
- 1920(H) x 1080(V) display
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable
- Commercial temperature range
- Landscape pin out. Portrait mode scan.
- Plastic substrate.

3. Mechanical specifications

Parameter	Specifications	Unit	Remark
Screen Size	10.8	Inch	
Display Resolution	1920(H) x 1080(V)	Pixel	
Active Area	239.04(H) x 134.46(V)	mm	
Pixel Pitch	0.1245(H) x 0.1245(V)	mm	
Pixel Configuration	Square		
Outline Dimension	247.04(H) x 153.76(V) x 0.439 (D)	mm	
Module Weight	22.5 +/- 2	g	
Number of Gray	16 Gray Level (monochrome)		
Display operating mode	Reflective mode		
Surface treatment	Non		

4. Mechanical drawing of EPD Module



5. Input/Output Terminals

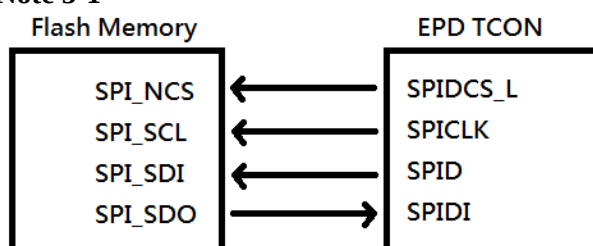
5-1) Pin out list

Golden finger design for: Aces 51607-0510M-001

Pin	Symbol	Description	Notes
1	VGL	Negative power supply gate driver	
2	VGH	Positive power supply gate driver	
3	VDD	Digital power supply drivers(3.3V)	3.3V
4	MODE	Output mode selection gate driver	
5	CKV	Clock Gate Driver	
6	VCOM2	Common voltage	Note 5-3
7	VCOM	Common voltage	Note 5-3
8	VDD	Digital power supply drivers(3.3V)	3.3V
9	VSS	Ground	
10	XCL	Clock source driver	
11	D0	Data signal source driver	
12	D1	Data signal source driver	
13	D2	Data signal source driver	
14	D3	Data signal source driver	
15	D4	Data signal source driver	
16	D5	Data signal source driver	
17	D6	Data signal source driver	
18	D7	Data signal source driver	
19	D8	Data signal source driver	
20	D9	Data signal source driver	
21	D10	Data signal source driver	
22	D11	Data signal source driver	
23	D12	Data signal source driver	
24	D13	Data signal source driver	
25	D14	Data signal source driver	
26	D15	Data signal source driver	
27	VSS	Ground	
28	XSTL	Start pulse source driver	Note 5-4
29	XLE	Latch enable source driver	
30	XOE	Output enable source driver	
31	XON	E Ink internal test pin	Note 5-2.
32	VPOS	Positive power supply source driver	

33	VNEG	Negative power supply source driver	
34	Border	Border connection	
35	NC	No Connection	
36	SHR_IN	Source Shift direction	H: S1920~S1 L: S1~S1920
37	STL1_IN	Source Start pulse input	SHR_IN H: output SHR_IN L: input
38	STL2_IN	Source Start pulse input	SHR_IN H: input SHR_IN L: output
39	VSS	Ground	
40	VSS	Ground	
41	UD_IN	Gate Shift direction	H:STV1→G1→G1080 L:STV2→G1080→G1
42	STV1_IN	Gate Start pulse input	UD_IN H: input UD_IN L: output
43	STV2_IN	Gate Start pulse input	UD_IN H: output UD_IN L: input
44	NC	No Connection	
45	VSS	Ground	
46	VDD2	SPI flash power supply (1.8V)	1.8V
47	SPI_SCL	Serial Data Clock for Flash memory	Note 5-1
48	SPI_NCS	Chip Select for Flash memory	Note 5-1
49	SPI_SDI	Serial Data Input for Flash memory	Note 5-1
50	SPI_SDO	Serial Data Output for Flash memory	Note 5-1
51	NC	No Connection	

Note 5-1



Note 5-2: Please connect to VDD voltage by 10K resistance.

Note 5-3: Connect to system board Vcom power..

Note 5-4: If SHR_IN = H, then XSTL connect to STL2_IN.

If SHR_IN = L, then XSTL connect to STL1_IN.

6. Electrical characteristics

6-1) Absolute maximum rating

Parameter	Symbol	Rating	Unit
Logic Supply Voltage	VDD	-0.3 to +5	V
SPI Supply Voltage	VDD2	-0.5 to +2.5	V
Positive Supply Voltage	V _{POS}	-0.3 to +18	V
Negative Supply Voltage	V _{NEG}	+0.3 to -18	V
Max .Drive Voltage Range	V _{POS} - V _{NEG}	36	V
Supply Voltage	VGH	-0.3 to VGL+50V	V
Supply Voltage	VGL	-25 to +0.3	V
Supply Range	VGH-VGL	+10 ~ +50	V
Operating Temp. Range	TOTR	0 to +50	°C
Storage Temperature	TSTG	-25 to +70	°C

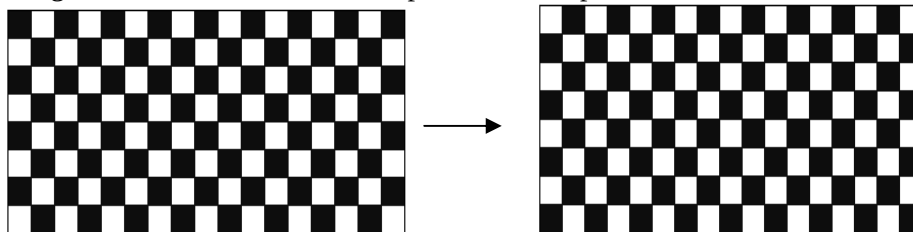
6-2) Panel DC characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Signal ground	V _{SS}		-	0	-	V
Logic Voltage supply	V _{DD}		3.0	3.3	3.6	V
	I _{VDD}	V _{DD} =3.3V	-	7.91	10.76	mA
SPI Voltage supply (Active)	V _{DD2}		1.65	1.8	1.95	V
	I _{VDD2}	V _{DD} =1.8V	-	1.69	-	mA
SPI Voltage supply (Standby)	V _{DD2}		1.65	1.8	1.95	V
	I _{VDD2}	V _{DD} =1.8V	-	0.04	-	mA
Gate Negative supply	V _{GL}		-20.5	-20	-20.5	V
	I _{GL}	V _{GL} = -20V	-	1.99	15.95	mA
Gate Positive supply	V _{GH}		27.5	28	28.5	V
	I _{GH}	V _{GH} = 28V	-	1.64	1.72	mA
Source Negative supply	V _{NEG}		-15.4	-15	-14.6	V
	I _{NEG}	V _{NEG} = -15V	-	0.5	91.08	mA
Source Positive supply	V _{POS}		14.6	15	15.4	V
	I _{POS}	V _{POS} = 15V	-	0.6	99.24	mA
Asymmetry source	V _{Asym}	V _{POS} +V _{NEG}	-800	0	800	mV
Common voltage	V _{COM}		-3.5	Adjusted	-1.5	V
	I _{COM}		-	0.2	-	mA
Power panel	P		-	131.4	3257.4	mW
Standby power panel	P _{STBY}		-	-	0.35	mW
Operating temperature			0	-	50	°C
Storage temperature			-25	-	70	°C

- The maximum average Currents for power consumption and Max. Currents are measured using 85 Hz waveform with following pattern transition: from black and white single checker pixel pattern to inversed black and white single checker pixel pattern. (Note 6-1)
- The Typical power consumption is measured using 85 Hz waveform with following pattern transition: from horizontal 4 gray scale pattern to vertical 4 gray scale pattern. (Note 6-2)
- The standby power is the consumed power when the panel controller is in standby mode.
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by E Ink.
- Vcom is recommended to be set in the range of assigned value ± 0.1 V

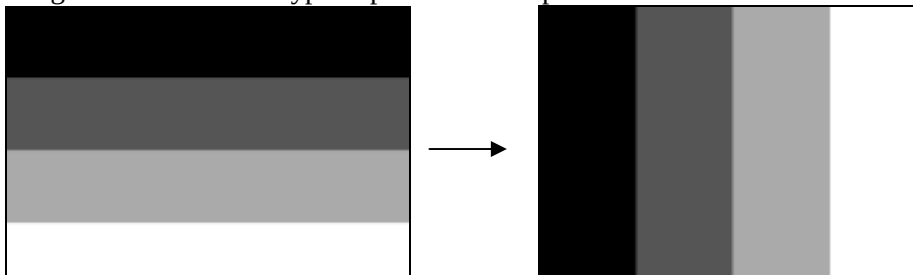
Note 6-1

Image flow to measure maximum power consumption.



Note 6-2

Image flow to measure typical power consumption.

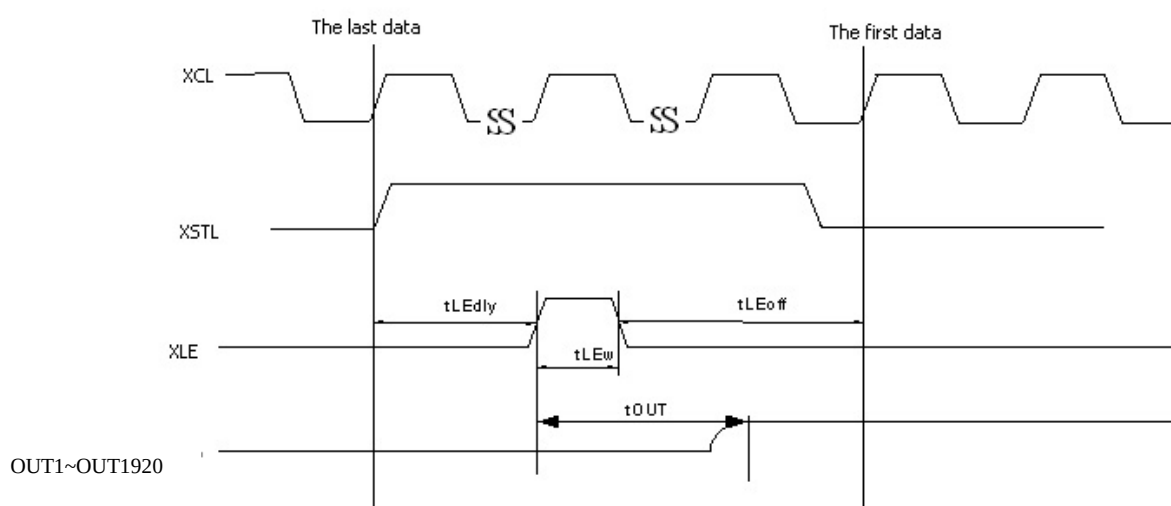


6-3)Panel AC characteristics

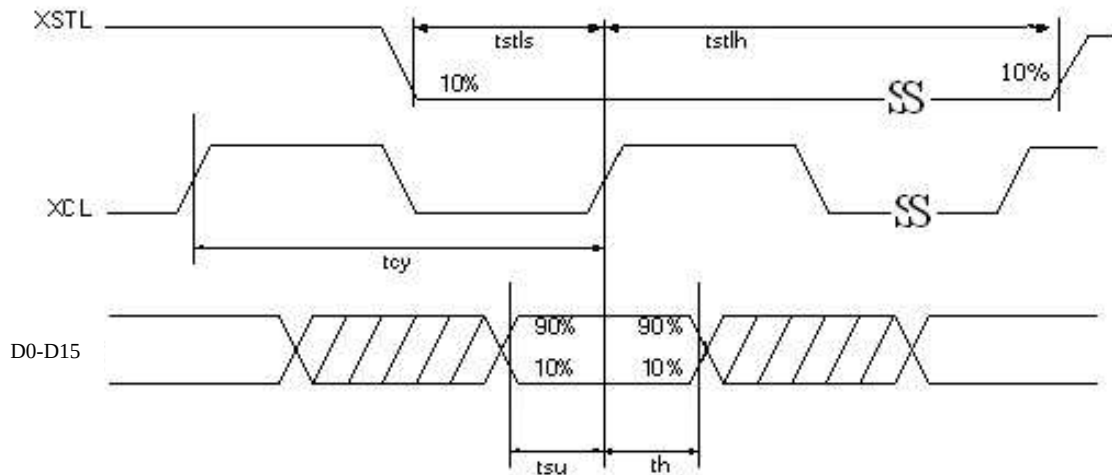
VDD=3.0V to 3.6V, unless otherwise specified.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Clock frequency	fckv	-	-	200	kHz
Minimum “L” clock pulse width	twL	500	-	-	ns
Minimum “H” clock pulse width	twH	500	-	-	ns
Clock rise time	trckv	-	-	100	ns
Clock fall time	tfckv	-	-	100	ns
SPV setup time	tSU	100	-	twH-100	ns
SPV hold time	tH	100	-	twH-100	ns
Pulse rise time	trspv	-	-	100	ns
Pulse fall time	tfspv	-	-	100	ns
Clock XCL cycle time	tcy	16.67	-	-	ns
D0 .. D7 setup time	tsu	8	-	-	ns
D0 .. D7 hold time	th	8	-	-	ns
XSTL setup time	tstls	0.5* tcy	-	0.8* tcy	ns
XSTL hold time	tstlh	0.5* tcy	-	-	ns
XLE on delay time	tLEdly	10.5*tCY	-	-	ns
XLE high-level pulse width (When VDD=1.7V to 2.1V)	tLEw	300	-	-	ns
XLE off delay time	tLEoff	200	-	-	ns
Output setting time to +/- 30mV(C _{load} =200pF)	tout	-	-	20	us

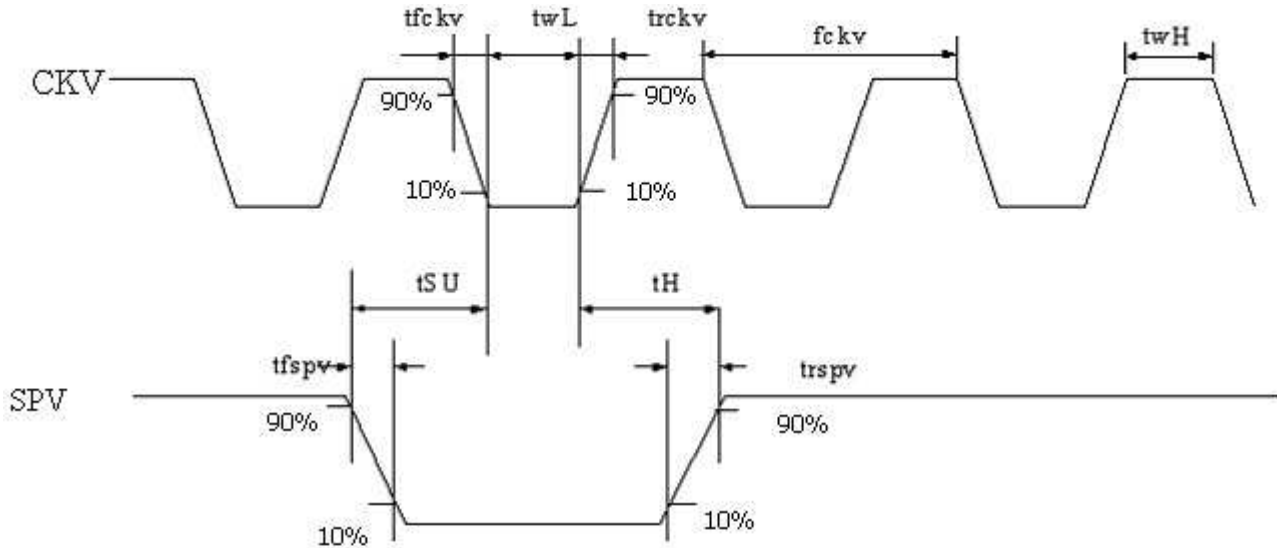
OUTPUT LATCH CONTROL SIGNALS



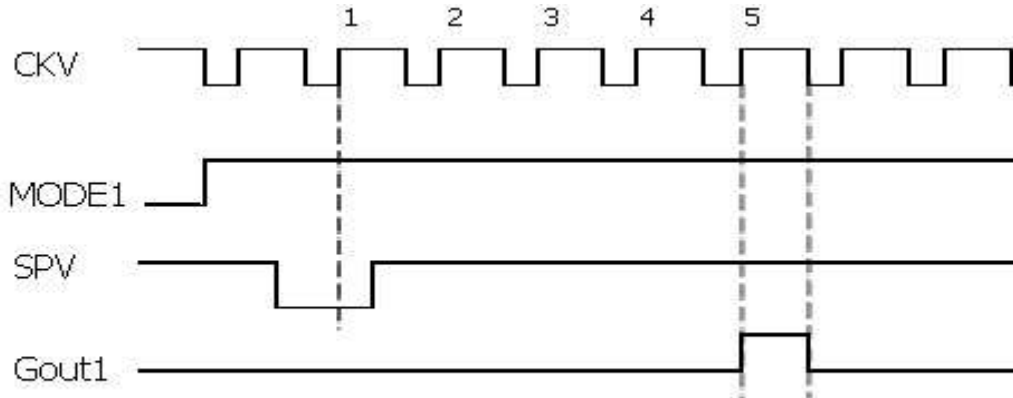
CLOCK & DATA TIMING



CKV & SPV TIMING



GATE OUTPUT TIMING



Note : First gate line on timing.

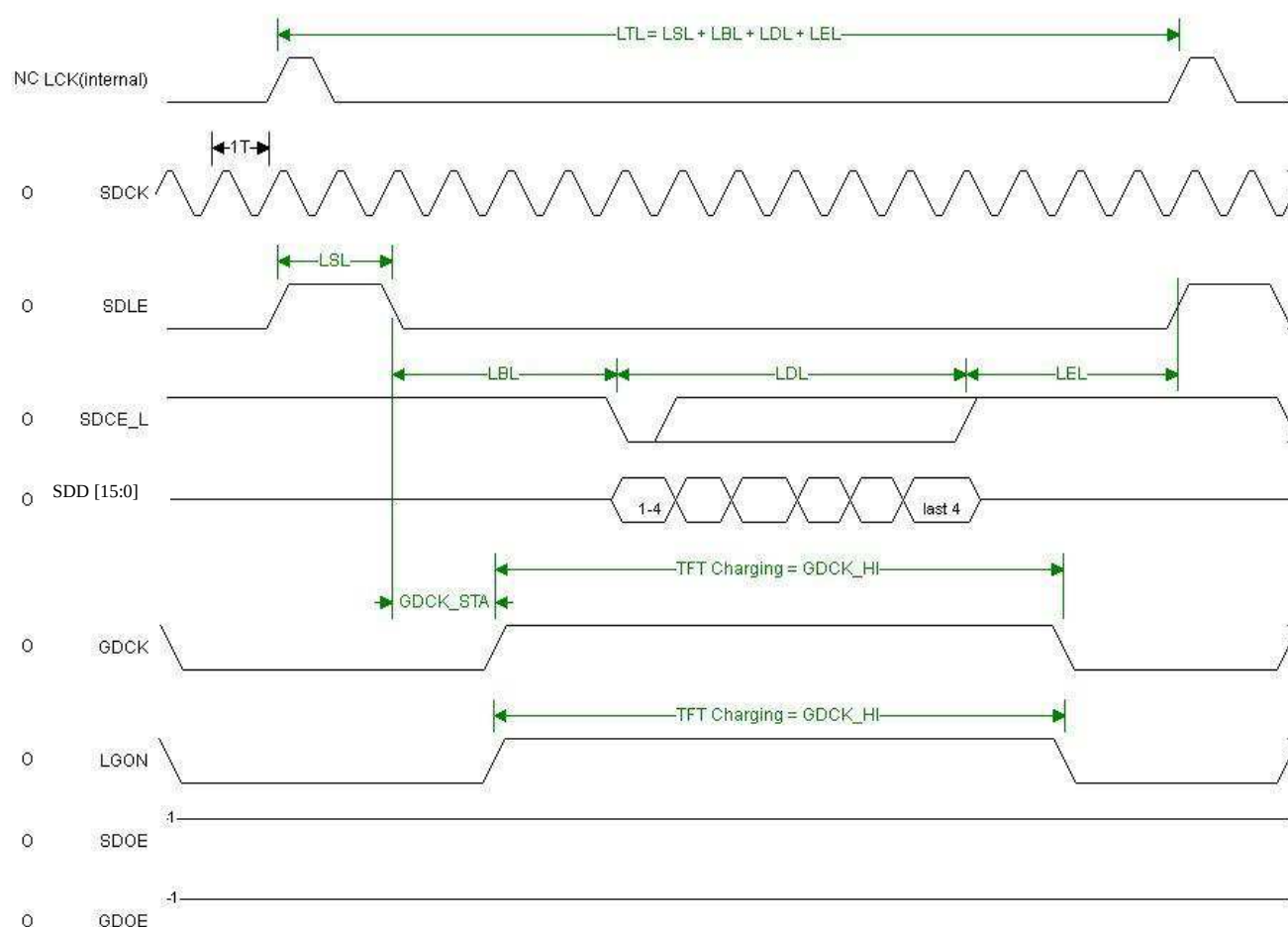
6-4)Refresh rate

The module is applied at a maximum refresh rate of 85 Hz.

	Min	Max
Refresh Rate	-	85 Hz

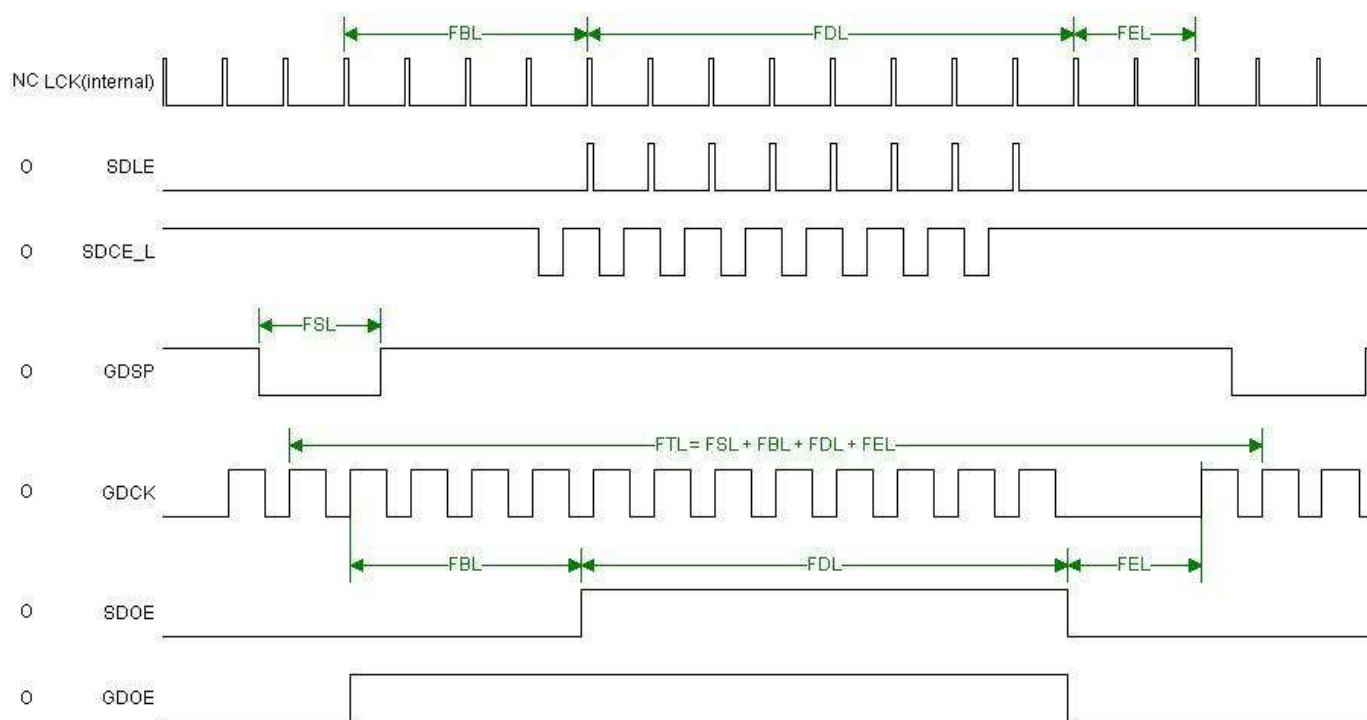
6-5) Controller Timing

This timing mode is depicted on Figure 1 and Figure 2 and it refers to timing of Source Driver Output Enable (SDOE) and Gate Driver Clock (GDCK). Note, that in this mode LGON follows GDCK timing.



Line Timing in Mode 3

Note 1: LCK is an internal signal and it is shown for reference only.



Frame Timing in Mode 3

Timing parameters table (SDCK= 33MHz)

Mode	3	Resolution 1920*1080				
SDCK[MHz]	33.00					
Pixels Per SDCK	8					
Line Parameters[SDCK]	LSL	LBL	LDL	LEL	GDCK_STA	LGONL
	10	7	240	96	2	264
Line Parameters[us]	-	-	-	-	-	-
	0.30	0.21	7.27	2.91	0.06	8.00
Frame Parameters[SDCK]	FSL	FBL	FDL	FEL	-	FR[Hz]
	1	4	1080	15	-	85
Frame Parameters[us]	-	-	-	-	-	-
	10.70	42.79	11553.00	160.45	-	-

Note 1: For parameters definition, see EPD panel timing.

Note 2: For Isis controller GDCK_STA and LGONL are not settable parameters; GDCK_STA = LBL, LGONL = LDL + 0.5

Note 3: For Freescale SOC GDOE low pulse represent FSL and GDSP pulses with the first period of FBL.

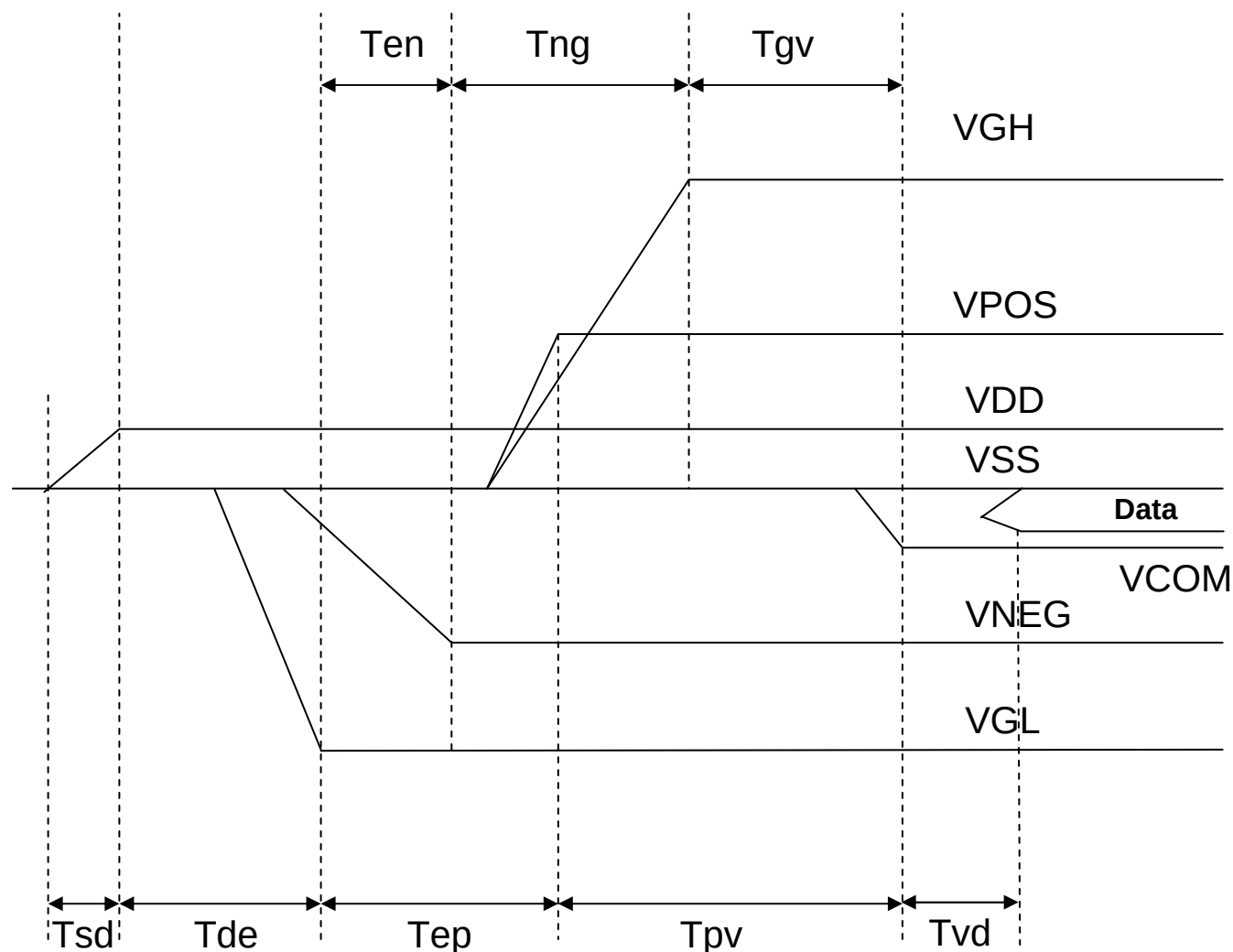
Note 4: SDOE = XOE, SDCE_L = XSTL, SDCK = XCL, GDCK = CKV, GDOE = MODE 1, GDSP = SPV

7. Power sequence

Power Rails must be sequenced in the following order :

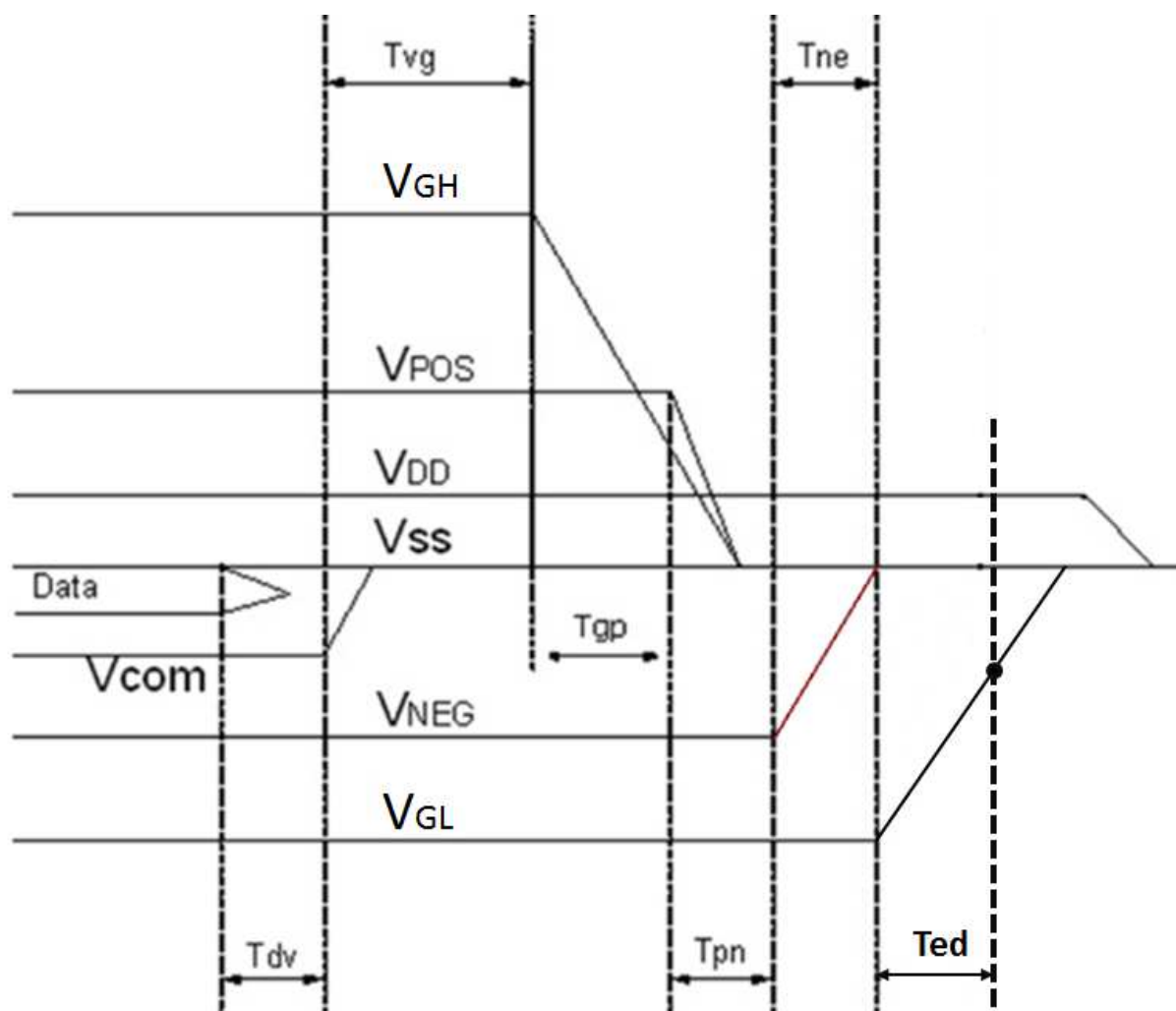
1. VSS → VDD → VNEG → VPOS (Source driver) → VCOM
2. VSS → VDD → VGL → VGH (Gate driver)

POWER ON



	Min	Max
Tsd	30us	-
Tde	100us	-
Tep	1000us	-
Tpv	100us	-
Tvd	100us	-
Ten	0us	-
Tng	1000us	-
Tgv	100us	-

POWER OFF



	Min	Max	Remark
Tdv	100μs	-	-
Tvg	0μs	-	-
Tgp	0μs	-	-
Tpn	0μs	-	-
Tne	0μs	-	-
Ted	0.5s	-	Discharged point @ -7.4 Volt

Note 7-1 : Supply voltages decay through pull-down resistors.

Note 7-2 : Begin to turn off VGL power after VNEG and VPOS are completely or almost discharged to GND state.

Note 7-3 : VGL must remain negative of Vcom during decay period

8. Optical characteristics

8-1) Specifications

Measurements are made with that the illumination is under an angle of 45 degrees, the detection is perpendicular unless otherwise specified.

T = 25°C

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT	Note
R	Reflectance	White	30	35	-	%	Note 9-1
Gn	N _{th} Grey Level	-	-	$DS + (WS - DS) \times n / (m - 1)$	-	L*	-
CR	Contrast Ratio	-	9	14	-		-
a*	Position between red/magenta and green	White	-3.95	-	0.95		Note 9-1
b*	position between yellow and blue	White	-2.55	-	1.95		Note 9-1

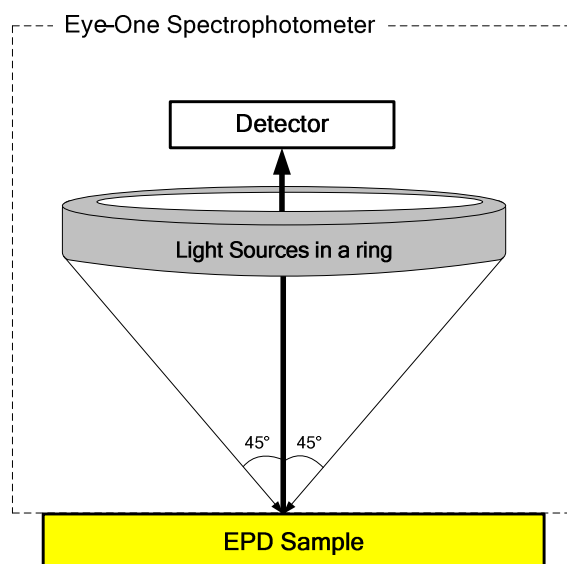
WS: White state, DS: Dark state, Gray state from Dark to White :DS、G1、G2...、Gn...、Gm-2、WS

m: 4、8、16 when 2、3、4 bits mode

Note 9-1: Spectrum meter: Eye – One Pro Spectrophotometer

8-2) Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd): $CR = Rl/Rd$



8-3) Reflection Ratio

The reflection ratio is expressed as:

$$R = \text{Reflectance Factor white board} \times (L_{\text{center}} / L_{\text{white board}})$$

L_{center} is the luminance measured at center in a white area (R=G=B=1). L_{white board} is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

9. Handling, safety and environment requirements
WARNING

The display TFT may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.

CAUTION

The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.

Disassembling the display module can cause permanent damage and invalidate the warranty agreements.

Observe general precautions that are common to handling delicate electronic components. The TFT can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

Data sheet status

Product specification	This data sheet contains final product specifications.
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Limiting values

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

Application information

Where application information is given, it is advisory and does not form part of the specification.

10. Reliability test

	TEST	CONDITION	METHOD
1	High-Temperature Operation	T = +50°C, RH = 30% for 72 hrs	IEC 60 068-2-2Be
2	Low-Temperature Operation	T = 0°C for 72 hrs	IEC 60 068-2-2Ae
3	High-Temperature Storage	T = +70°C, RH=40% for 72 hrs (Test In White Pattern)	IEC 60 068-2-2Bb
4	Low-Temperature Storage	T = -25°C for 72 hrs (Test In White Pattern)	IEC 60 068-2-1Ab
5	High-Temperature, High-Humidity Operation	T = +40°C, RH = 90% for 72 hrs	IEC 60 068-2-78
6	High Temperature, High- Humidity Storage	T = +60°C, RH=80% for 72 hrs (Test In White Pattern)	IEC 60 068-2-78
7	Temperature Cycle	-25°C → +70°C, 50 Cycles 30mins 30 mins (Test In White Pattern)	IEC 60 068-2-14Nb
8	Solar Radiation test	765 W/m ² for 72hrs, 40°C (Test In White Pattern)	IEC60 068-2-5Sa
9	Package Vibration	1.04G, Frequency: 10~500Hz Direction: X,Y,Z Duration: 1 hours in each direction	Full packed for shipment
10	Package Drop Impact	Drop from height of 100 cm on concrete surface. Drop sequence: 1 corner, 3 edges, 6 faces One drop for each.	Full packed for shipment
11	Electrostatic Effect (non-operating)	(Air mode)+/-12kV; (Five point test in active area.) (Contact mode)+/-8kV (Five point test in active area.)	IEC61000-4-2

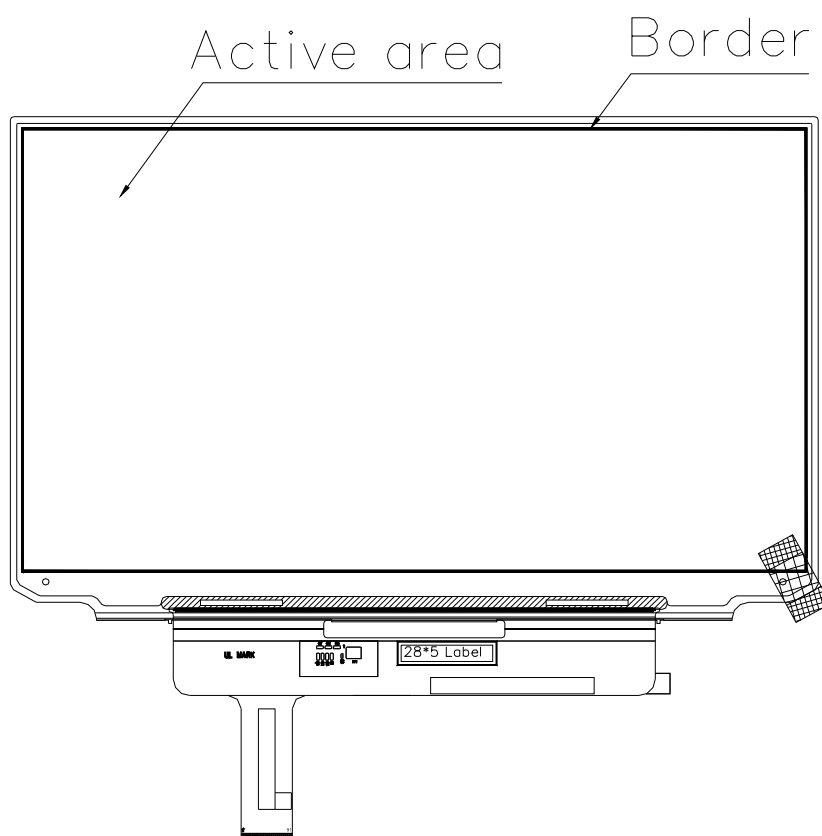
Actual EMC level to be measured on customer application

Note : The protective film must be removed before temperature test.

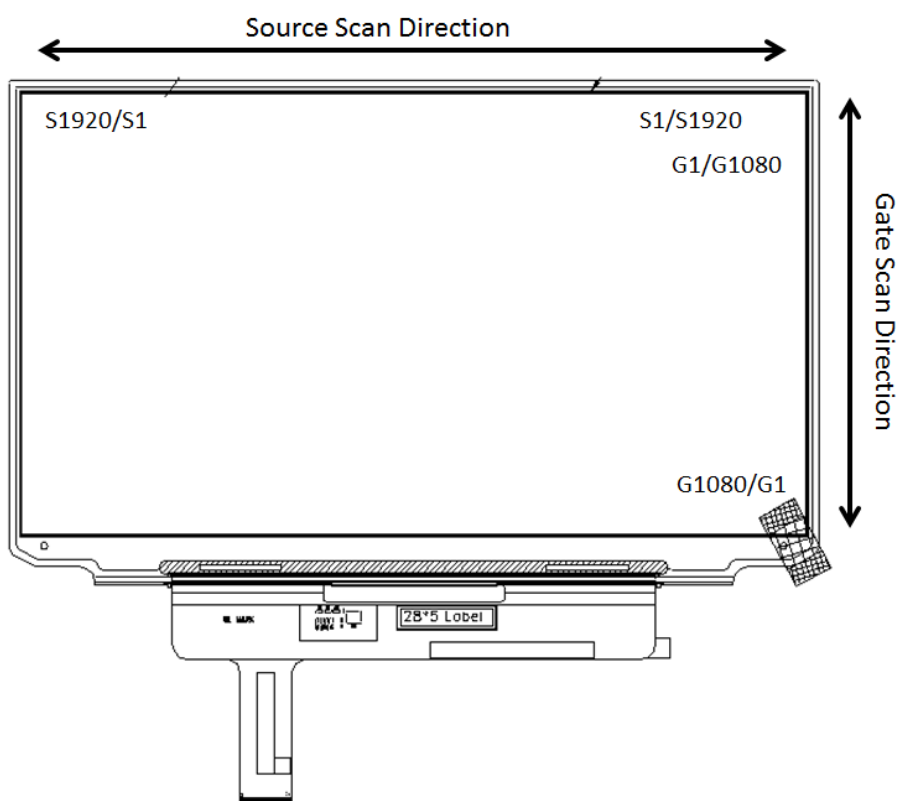
[Criteria]

In the standard conditions, there is not display function NG issue occurred.
(Including : line defect, no image) All the cosmetic specification is judged before the reliability stress.

11. Border definition and scan direction

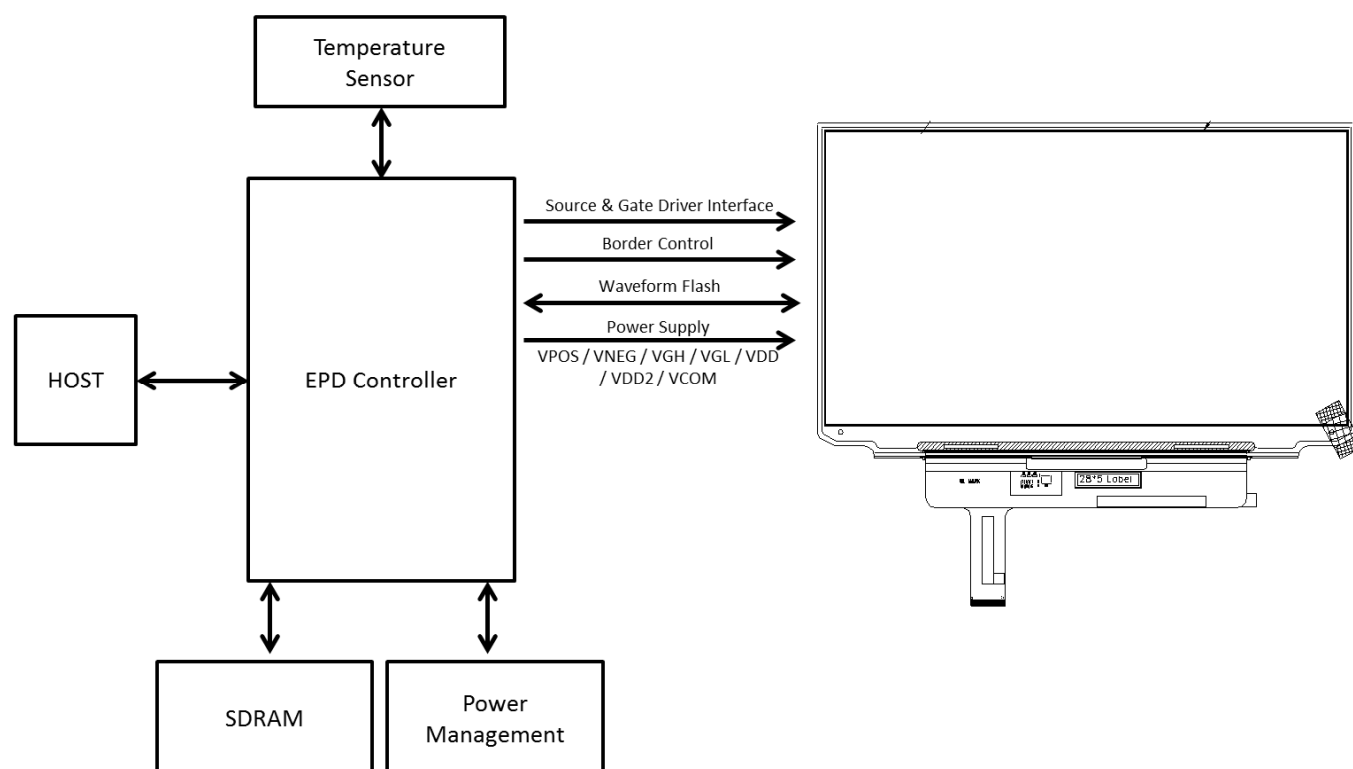


Border definition



Scan direction

12. Connecting diagram



13.Packing

REV	DESCRIPTION	DESIGN	DATE
01	INITIAL RELEASE	Maggie Chen	20190725
02	增加出貨單內容, Tray 與 出貨料號 Label 卡位置 (ITEM 8)	Maggie Chen	20191202

NOTE:

- One layer include: 2 pcs panel, 1 pc epe cushion sheet & 1 pc tray
- Q'TY: 24 pcs panel/carton.
- Dimension: 558*370*170mm
- Weight: TBD
- Tray 需 180°堆疊, 堆疊後可從側邊檢視圓弧防呆方向是否正確
- 靜電袋封好之後, 在上方附上 24pcs Barcode Label

8	Barcode Label (25mm*8mm)	24	
7	EPE CUSHION SHEET	12	抗靜電
6	30g 加厚複合紙/活版乾燥劑 73*95mm(禹科JK0030)	2	
5	CARTON INTERNAL	1	
4	FOLDED BAG, OTHER(562*393*593)	1	抗靜電
3	ES108FC2	24	
2	TRAY	13	抗靜電
1	EPE FOAM	2	
ITEM	DESCRIPTION	Q'TY	REMARK

M.TL.SPBC.	UNSPECIFIED TOL'S ±5.0mm	REMARK	元太科技工業股份有限公司 E Ink Holdings Inc.
APPROVE	Kevin Cheng	SCALE 1:1	DWG.TITLE ES108FC2 PACKING
CHECK	Kevin Cheng	UNIT mm	
DESIGN	Maggie Chen	SHEET 1 OF 1	
MTL.NO.		DWG.NO.	
		A4 SIZE	